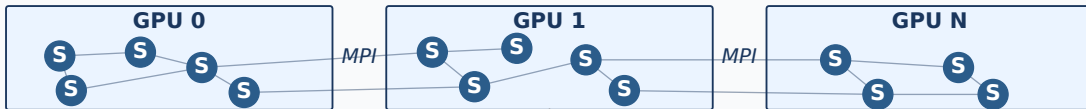


Two-level parallelism on an Exascale HPC system

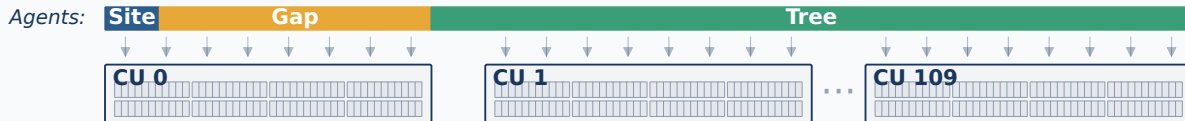
Level 1: Inter-rank MPI — site's ensemble colocated (1 rank/GPU)



Per colocated site: 1 site agent + gap agents (e.g. 500/site) + tree agents (e.g. 1000/gap)

zoom into one GPU

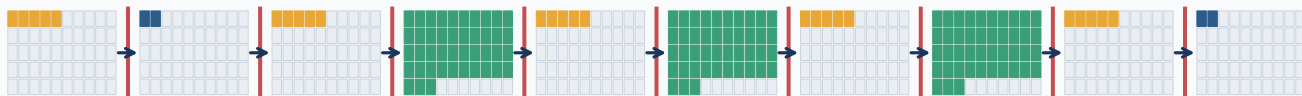
Level 2: Intra-rank GPU threads — 1 agent = 1 thread (grid-stride at scale)



AMD MI250X GCD: 110 CUs · 8 blocks/CU · 128 threads/block

Multi-priority kernel pipeline (inside each tick)

Fused kernel per tick: 10 priorities in order → 1 launch via on-device grid barriers



each cell = 1 thread ■ active (Site/Gap/Tree) □ idle at barrier |=grid barrier